Co-Wingman

A Hybrid Edge-Cloud AI Assistant for Real-Time Conversational Recovery Cues

Mrunal Vibhute, Shrimayee Deshpande, Honey Reddy Daram, Dr. Andrea Ramirez-Salgado
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01 Motivation

Help speakers recover their train of thought with a short, context-aware cue during a conversational pause.

02 Capabilities

01 Local speech recognition
02 60-second rolling context
03 Manual or pause triggers
04 Portfolio-grounded cues
05 Minute summaries
06 Optional S3 continuity

03 Privacy & control

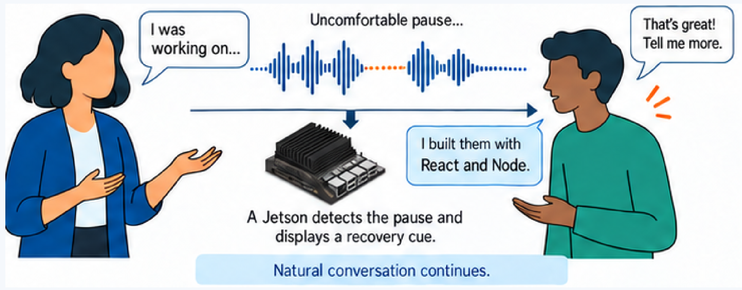
Audio stays local
No application-created recordings.

Bounded cloud input
Recent text goes to UF NaviGator.

Storage is opt-in
Saved S3 objects persist until deleted.

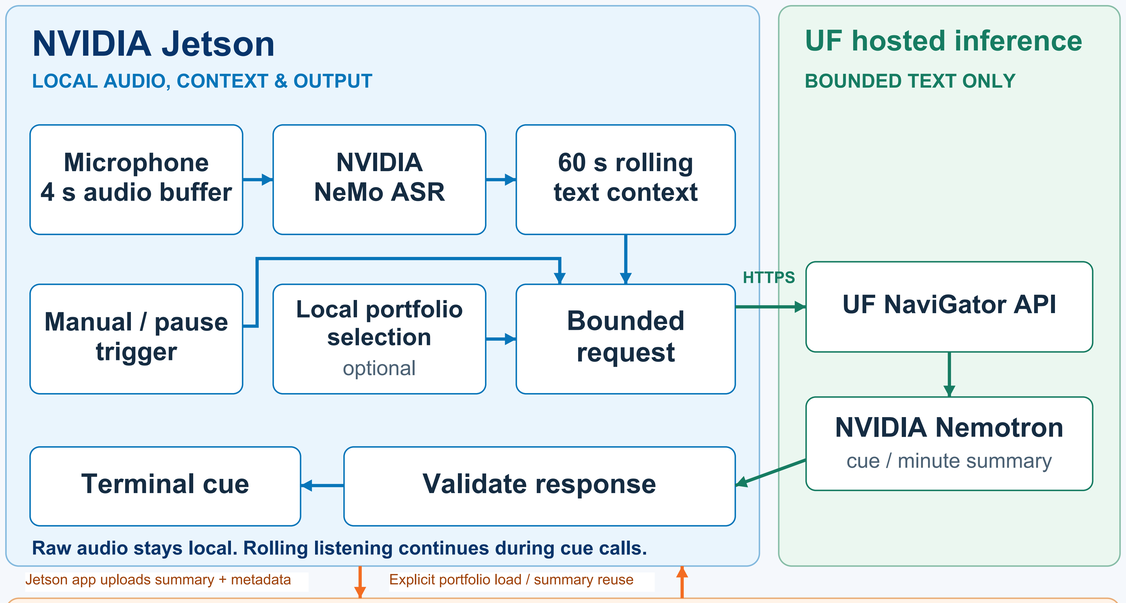
User-controlled assistance
Pause cues use an energy/silence heuristic, not anxiety detection.

04 Speak. Pause. Recover.

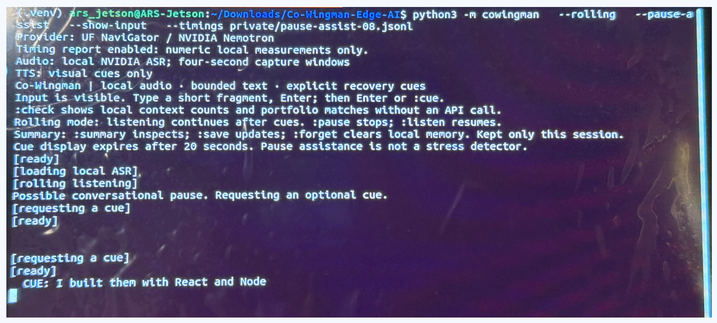


A brief cue helps the speaker continue the conversation.

05 Hybrid architecture



06 Results & next steps



Live visual cue demonstrated on Jetson

CUE: I built them with React and Node

94	13	0
audio windows	cue requests	dropped windows

Measured stage	Median	p95
ASR inference	110 ms	127 ms
Cue handling	1.04 s	2.54 s
Summary update	1.42 s	1.99 s

Timing sample: pause-assist-05; photo from a separate demo. 69 voiced windows, 7 summaries, zero recorded stage failures. Cue timing excludes capture wait; a completed request may offer no cue.

NEXT STEPS

01 Reduce four-second buffering delay
02 Evaluate cue usefulness with users
03 Validate AWS upload and recovery
04 Evaluate experimental disfluency scoring

Local speech processing / Bounded cloud reasoning / Optional persistent memory

github.com/Co-Wingman-AI/Co-Wingman-Edge-AI